

Orhan Kaplan, Ph.D.

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PROFESSIONAL SUMMARY

Senior Data Scientist specializing in causal inference and predictive behavioral/psychological modeling, with nearly 10 years of R&D experience including a graduate research assistantship at the University of Florida, and a record of peer-reviewed publications. Skilled at turning messy, large-scale user data into rigorous program and impact evaluations — designing studies, building statistical frameworks that isolate true feature impact where standard A/B testing is inapplicable, and delivering impactful findings into clear decision-making insights for stakeholders across industries.

TECHNICAL SKILLS

Data Science/Applied Research:

- Hypothesis Testing, Experimental/Quasi-Experimental Observational Studies Design, Structural Equation Modeling, Propensity Score Matching, Entropy Balancing, Fixed-Effects Modeling, Machine Learning (Multiple Regression, Lasso & Ridge Regression, Logistic Regression, kNN Regression/Classification, Decision-Tree ensembles), Multiple imputation (MI, MICE), and Qualitative Research.

Programming & Development:

- **Languages & Tools:** Python, SPSS, Stata, MPlus, Git, SQL (SQLiteStudio, MySQL).
- **Machine Learning:** scikit-learn, cross-validation, classification, Random Forest, XGBoost.
- **Data & Visualization:** pandas, NumPy, Stata, matplotlib/seaborn, streamlit.

Communication & Collaboration:

- Strong in rigorous academic/technical writing, team collaboration, supporting a productive and inclusive environment.

Languages:

- Fluent in English and Turkish.

PROFESSIONAL EXPERIENCE

Lecturer | Gaziantep University,

April 2019 – Present

- Designed/analyzed data for research projects using SPSS/Stata/Mplus/Dedoose/Python
- Conducted independent and collaborative research on psychological, educational, behavioral factors, resulting in 10 peer-reviewed publications
- Taught courses on research methodology at graduate level (delivered in English) and undergraduate level, mathematics, teaching methods and vocational English.
- Collaborated on research projects, social awareness projects, and building a Mathematics Library

Graduate/OPS Research Assistant | Higher Education Department, University of Florida, *October 2016 – May 2018*

- Managed and cleaned large-scale datasets
- Screened datasets and conducted statistical analyses using Stata (Regression-Discontinuity Design, Fixed-effects Modeling, Logistic Regression)
- Screened state-level educational policies and literature review

Mathematics Instructor | Upward Bound Program, University of Florida,

January 2012 – May 2012

- Instructed high school students in ACT-oriented geometry.
- College preparatory program for students in poverty.

Mathematics & Science Teacher | Batman, Turkey,

September 2006 – February 2008

- İkiztepe Middle School (2006-2008) and Şehit Öğretmenler Middle School (2007-2008)

SELECTED PROJECTS

MICE-PMM Diagnostics Toolkit: Open-source Python package implementing MICE with Predictive Mean Matching (Rubin, 1987; van Buuren, 2018) for complex survey data, with a diagnostics suite (Rubin's Rules, FMI, RVI, RE, Von Hippel's recommended-M, convergence plots) not previously available together in existing open-source tools.

Telco Customer Churn Analysis & Predictive Modeling Pipeline: End-to-end pipeline identifying high-risk churn segments and simulating ROI of retention strategies using logistic regression and KNN classification, with custom data quality checks and multi-model ROC evaluation.

Sci-Fi Movie Recommendation System Using KNN Collaborative Filtering: Built a content recommendation engine using distance-based KNN regression, optimized with Manhattan distance and validated via a 75/25 train-test split.

Applications of AI in Mathematics Instruction | TÜBİTAK Grant 222B158 (2023): Instructed mathematics educators on integrating AI-driven tools into instructional design, as part of a nationally funded technological innovation initiative for math education.

EDUCATION

Ph.D. in Curriculum & Instruction, University of Florida, FL, USA 2018

- Specialization in Mathematics Education and Research and Evaluation Methodology

M.S.E. in Mathematics, Arkansas State University, AR, USA 2011

- Thesis on *Stone-Weierstrass Approximation Theorem*

Studying Abroad Scholarship, Ministry of National Education, Turkey October 2007

- A selective and comprehensive scholarship covering education for Master's and PhD degrees

CERTIFICATIONS

Machine Learning and AI, HarvardX, June 2026

Machine Learning and AI Foundations: Causal Inference and Modeling, LinkedIn Learning, June 2026

Learning Git and GitHub, LinkedIn Learning, June 2026

Introduction to Data Science with Python, HarvardX, May 2026

SELECTED PUBLICATIONS & PRESENTATIONS

Kaplan, O. (2026). Cross-cultural control–value mechanisms on the detrimental effect of bullying on mathematics anxiety. *Behavioral Sciences*, 16(1), Article 3. <https://doi.org/10.3390/bs16010003>

Aydın Gürlü, S. & Kaplan, O. (2026). The effect of school bullying on science achievement and cognitive appraisals. *Problems of Education in the 21st Century*, 84(1), 7–26. <https://doi.org/10.33225/pec/26.84.07>

Kaplan, O., Kacar, T. & Simsek, N. (2025). The effect of bullying on epistemic emotions using Propensity Score Matching: Examining cross-national differences. The paper was presented at 2025 European Conference on Educational Research, Belgrade, Serbia.

Kaplan, O., Kara, M., & Kaçar, T. (2025). The effect of digital technology integration on students' creative thinking skills: PISA evidence using propensity score matching. *Problems of Education in the 21st Century*, 83(3), 403–423. <https://doi.org/10.33225/pec/25.83.403>

Aydın Gürlü, S. & Kaplan, O. (2023). Attitudes towards STEAM, critical thinking disposition and decision-making skills: Mediation and gender moderation. *International Journal of Contemporary Educational Research*, 10(1), 210–223. <https://doi.org/10.33200/ijcer.1272051>

Full publication list available at <https://orcid.org/0000-0002-4306-2266> or <https://nalpako2027.github.io/about/>